28/09/2024
Para. Inf.

X having density $f(x|\theta): \theta \in \Theta$. Test $H_0: \theta = \theta_0$ vs. $H_1: \theta = \theta_1$. NP Lemma $\Rightarrow$

The test $\phi(x) = \begin{cases} 1 & \text{if } f(x|\theta_1) > K f(x|\theta_0) \\ 0 & \phantom{\text{if } f(x|\theta_1)} < \end{cases}$ when $K \geq 0$, $\mathbb{E}_{\theta_0}\phi(x) = \alpha$

is MP ~~size~~-α test

<u>Note</u>: MP test from NP Lemma is always based on suff stat $T(\underset{\sim}{X})$ b/c.

$$\frac{f(x|\theta_1)}{f(x|\theta_0)} = \frac{h(x)\, g_{\theta_1}(T(x))}{h(x)\, g_{\theta_0}(T(x))} = \psi(T(x))$$

<u>Ex.</u> $X_1, \dots, X_n \overset{iid}{\sim} N(0, \sigma^2)$. Test $H_0: \sigma = \sigma_0$ (known) vs. $H_1: \sigma = \sigma_1$.
Find MP ~~size~~-α test (suppose $\sigma_1 > \sigma_0$)

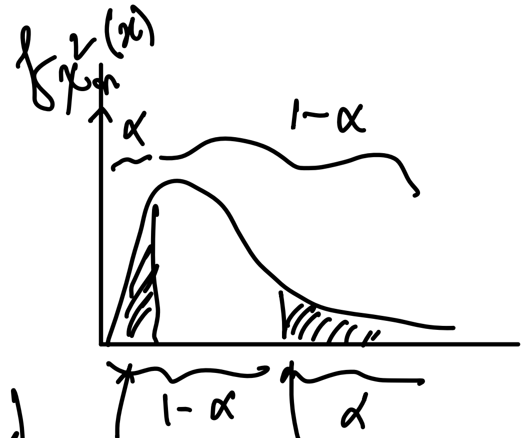
$$\lambda(\underset{\sim}{x}) = \frac{f(\underset{\sim}{x}|\sigma_1)}{f(\underset{\sim}{x}|\sigma_0)} = \left(\frac{\sigma_0}{\sigma_1}\right)^n e^{-\frac{1}{2}\sum_i x_i^2 \left(\frac{1}{\sigma_1^2} - \frac{1}{\sigma_0^2}\right)} \uparrow \quad T(x) = \sum_{i=1}^n x_i^2$$

rejection region $R = \{\underset{\sim}{x} : \lambda(\underset{\sim}{x}) > K\} = \{\underset{\sim}{x} : \sum_{i=1}^n x_i^2 > c\}$

Find c : $\mathbb{P}_{\sigma_0}\left(\sum_{i=1}^n X_i^2 > c\right) = \alpha$

$$\Rightarrow \mathbb{P}_{\sigma_0}\left(\chi^2_n > \frac{c}{\sigma_0^2}\right) = \alpha$$

$$\Rightarrow c = \sigma_0^2 \chi^2_{n,\alpha}$$



$\chi^2_{n,1-\alpha}$ $\quad$ $\chi^2_{n,\alpha}$ = upper α-th quantile

The MP size α test

$$\phi(\underset{\sim}{x}) = \mathbb{1}\left(\sum_{i=1}^n x_i^2 > \sigma_0^2 \chi^2_{n,\alpha}\right)$$

If $\underline{\sigma_1 < \sigma_0}$: $\lambda(x) \downarrow$ $T(x) = \sum_{i=1}^n x_i^2$

$$\mathbb{P}_{\sigma_0}\left(\sum_{i=1}^n X_i^2 > c\right) = \alpha$$

$$\phi(\underset{\sim}{x}) = \mathbb{1}\left(\sum_{i=1}^n x_i^2 < \sigma_0^2 \chi^2_{n,1-\alpha}\right) \Rightarrow c = \sigma_0^2 \chi^2_{n,1-\alpha}$$

Ex. $X_1, \dots, X_n \overset{iid}{\sim} U(0,\theta)$. Test $H_0 : \theta = \theta_0$ vs. $H_1 : \theta = \theta_1$

Find MP Size-α test (Suppose $\theta_1 > \theta_0$)

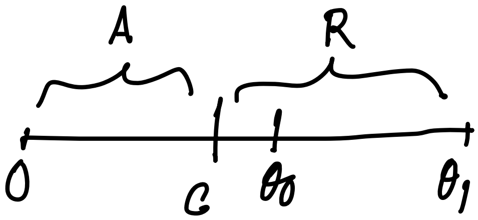
$$\frac{f(x|\theta_1)}{f(x|\theta_0)} = \left(\frac{\theta_0}{\theta_1}\right)^n \frac{1(0 \le x_{(n)} \le \theta_1)}{1(0 \le x_{(n)} \le \theta_0)} = \begin{cases} \left(\frac{\theta_0}{\theta_1}\right)^n & \text{if } 0 \le x_{(n)} < \theta_0 \\ \infty & \text{if } \theta_0 < x_{(n)} \le \theta_1 \end{cases}$$

We defined $\frac{1}{0}$ as ∞ b/c that ensures rejection of H_0 with prob 1 when $x_{(n)} \in (\theta_0, \theta_1)$

$\therefore R = \{\underset{\sim}{x} : x_{(n)} > c\}$. Find $c : \mathbb{P}_{\theta_0}(X_{(n)} > c) = \alpha$

$$f_{X_{(n)}}(x) = \frac{n}{\theta^n} x^{n-1} 1(0 < x < \theta)$$

$$\int_c^\infty \frac{n x^{n-1}}{\theta_0^n} 1(0 < x < \theta_0)\, dx = \alpha$$



$$\Rightarrow \int_c^{\theta_0} \frac{n x^{n-1}}{\theta_0^n} dx = \alpha$$

$$\Rightarrow \frac{n}{\theta_0^n} \frac{x^n}{n}\Big|_c^{\theta_0} = \alpha \Rightarrow 1 - \left(\frac{c}{\theta_0}\right)^n = \alpha$$

$$\Rightarrow c = (1-\alpha)^{1/n} \theta_0$$

Remark: We did not say anything about if

$$x \in B := \left\{x : \frac{f(x|\theta_1)}{f(x|\theta_0)} = K\right\}$$

If $\mathbb{P}(B) > 0$, then we may need to randomize on to get a size-α test

We may have to consider a randomized test

$$\phi(\underset{\sim}{x}) = \begin{cases} 1 & \text{if } f(x|\theta_1) > K f(x|\theta_0) \\ \gamma & \text{if } \quad = \\ 0 & \text{if } \quad < \end{cases}$$

where $\gamma \in [0,1]$ represents the prob. with which we reject H_0 when $x \in B$. We need to choose K and γ:

$$\mathbb{E}_{H_0}[\phi(X)] = \alpha \iff \mathbb{P}_{\theta_0}(f(X|\theta_1) > K f(X|\theta_0)) + \gamma \mathbb{P}_{\theta}(f(X|\theta_1) = K f(X|\theta_0)) = \alpha$$

<u>Note:</u> $S_K = \left\{x : \frac{f(x|\theta_1)}{f(x|\theta_0)} > K\right\} \downarrow K$ We will choose smallest $\underset{= K_0}{k}$:

$$\mathbb{P}_{\theta_0}\left(\frac{f(X|\theta_1)}{f(X|\theta_0)} > K\right) < \alpha$$

For that $K = K_0$, solve $\mathbb{P}_{\theta_0}\left(\frac{f(X|\theta_1)}{f(X|\theta_0)} > K_0\right) + \gamma \mathbb{P}_{\theta_0}\left(\frac{f(X|\theta_1)}{f(X|\theta_0)} = K_0\right) = \alpha$

$X \sim f \quad H_0 : f = f_0$ vs. $H_1 : f = f_1$

<u>Ex.</u>

x	1	2	3	4	5	6
$f_0(x)$	0.01	0.01	0.01	0.01	0.06	0.9
$f_1(x)$	0.05	0.04	0.03	0.02	0.06	0.8
$\lambda(x)$	5	4	3	2	1	8/9

Find MP size-α test for $\begin{cases} \alpha = 0.05 \\ \alpha = 0.03 \end{cases}$

$\lambda(x) > K$

$\lambda(x) \downarrow x$

NP Lemma $\Rightarrow$ reject H_0 if $X < c$

$P_{f_0}(X < c)$

Find largest c

So $c = 5$ $\quad \phi_1(X) = \begin{cases} 1 & \text{if } X < 5 \\ 0 & \quad \geq 5 \end{cases}$

$$\mathbb{E}_{f_0}\phi_1 = 0.04 < 0.05$$

$\because$ ϕ_1 is a level-α, but not size-α

So we need to consider a randomized test to get size-α

For $c=5$

$$\phi(x) = \begin{cases} 1 & \text{if } x<5 \\ \gamma & \text{if } x=5 \\ 0 & \text{if } x>5 \end{cases} = \begin{cases} 1 & \text{if } x<5 \\ 1/6 & \text{if } x=5 \\ 0 & \text{if } x>5 \end{cases}$$

$$P_{f_0}(X<5) + \gamma P_{f_0}(X=5) = 0.05$$

$$\Rightarrow \gamma = \frac{1}{6}$$

NP Lemma (full statement):

Suppose $X \sim f_\theta$; $\theta \in \Theta$. Test $H_0: \theta=\theta_0$ vs. $H_1: \theta=\theta_1$

(i.e. $f_{\theta_0} \equiv f_0$, $f_{\theta_1} \equiv f_1$) $\quad f = f_0$ (known) $\quad f = f_1$ (known)

(1) [Existence] For prefixed $0<\alpha<1$, $\exists$ a MP size-α test given by

$$\phi^*(x) = \begin{cases} 1 & \text{if } f_1(x) > K f_0(x) \\ \gamma & \text{if } f_1(x) = K f_0(x) \\ 0 & \text{if } f_1(x) < K f_0(x) \end{cases}$$

where $K(\geq 0)$ and $\gamma \in [0,1]$ is determined such that $\mathbb{E}_0[\phi^*(X)] = \alpha$ [E_0, E_1 are expectations under f_0, f_1; P_0, P_1 " prob. " " "]

(2) [Uniqueness (almost)] If ϕ_0 is a MP level-α test then ϕ_0 must satisfy

$$\phi_0(x) = \begin{cases} 1 & \text{if } f_1(x) > K f_0(x) \\ 0 & \text{if } f_1(x) < K f_0(x) \end{cases} \quad \mu \text{ a.e.}$$

for some $K \geq 0$

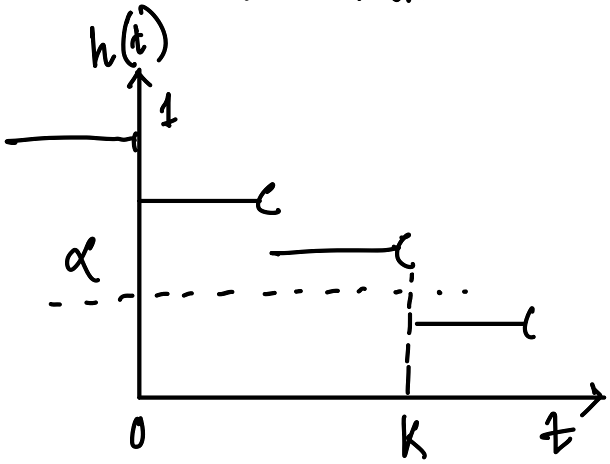
(i.e. $\mathbb{P}_i(\phi_0 \neq \text{given values}) = 0 \quad \forall\, i=0,1$)

Pf: (1) Define: $h(t) := \mathbb{P}_0\left(f_1(X) > t f_0(X)\right)$

$$= \mathbb{P}_0\left(\frac{f_1(X)}{f_0(X)} > t\right)$$

$$= 1 - F_0(t) \longrightarrow \text{CDF of } \frac{f_1(X)}{f_0(X)}$$

is a non-increasing in t



$\exists\, k \geq 0 : h(k-) \geq \alpha \geq h(k)$

Note: $h(k-) - h(k)$

$$= \mathbb{P}_0\left(f_1(X) = k f_0(X)\right)$$

Define,

$$\gamma = \begin{cases} \dfrac{\alpha - h(k)}{h(k-) - h(k)} & \text{if } h(k) \neq h(k-) \\ 0 & \text{if } = \end{cases}$$

$$\mathbb{E}_0 \phi^* = \underbrace{\mathbb{P}_0\left(f_1(X) > k f_0(X)\right)}_{h(k)} + \gamma\, \mathbb{P}_0\left(f_1(X) = k f_0(X)\right)$$

$$= h(k) + \frac{\alpha - h(k)}{h(k-) - h(k)}\left(h(k-) - h(k)\right) = \alpha$$

existence of k, γ are done

To show if ϕ any test: $\mathbb{E}_0 \phi \leq \alpha$ (or $= \alpha$)

then $\mathbb{E}_1 \phi^* \geq \mathbb{E}_1 \phi$

Suppose $A = \{\underset{\sim}{x} : \phi^*(x) \neq \phi(x)\}$. If $x \in A$, $\underset{>0}{\phi^*(x) - \phi(x)}$

$$\Rightarrow f_1(x) \geq k f_0(x)$$

$\left(\text{b/c if } f_1(x) < K f_0(x) \Rightarrow \phi^*(x) = 0,\ \phi(x) > 0 \Rightarrow \phi^* - \phi > 0\right)$

If $x \in A$ and $\phi^*(x) - \phi(x) < 0 \Rightarrow f_1(x) \leq K f_0(x)$

$\left(\text{b/c if } f_1 > K f_0 \Rightarrow \phi^* = 1,\ \phi < 1 \Rightarrow \phi^* - \phi > 0\right)$

$$\therefore\ g(x) = \left(\phi^*(x) - \phi(x)\right)\left(f_1(x) - K f_0(x)\right) \geq 0$$

$$\Rightarrow \int_A g(x)\,dx = \int_{\mathbb{R}^n} g(x)\,dx \geq 0$$

$$\Longrightarrow \int_{\mathbb{R}^n} \left(\phi^*(x) - \phi(x)\right) f_1(x)\,dx \geq K \int_{\mathbb{R}^n} \left(\phi^*(x) - \phi(x)\right) f_0(x)\,dx$$

$$\mathbb{E}_1\phi^* - \mathbb{E}_1\phi \geq K\left(\underbrace{\mathbb{E}_0\phi^*}_{=\alpha} - \underbrace{\mathbb{E}_0\phi}_{\leq\alpha}\right)$$

$$\geq 0$$

$$\therefore\ \mathbb{E}_1\phi^* \geq \mathbb{E}_1\phi$$

<u>Pf (Uniqueness):</u> Consider $B := \{x : \phi^*(x) \neq \phi_0(x),\ f_1(x) \neq K f_0(x)\}$

MP size-α test from NP Lemma

$$B^c = \underbrace{\{x : \phi^*(x) = \phi_0(x)\}}_{B_1} \cup \underbrace{\{f_1(x) = K f_0(x)\}}_{B_2}$$

$$= B_1 \cup B_2$$

If $x \in B$, $f_1(x) > K f_0(x) \Rightarrow \phi^*(x) = 1,\ \phi_0(x) < 1$

$\Rightarrow \phi^*(x) - \phi_0(x) > 0$

if $x \in B$, $f_1(x) < K f_0(x) \Rightarrow \phi^*(x) = 0$, $\phi_0(x) > 0$

$$\Rightarrow \phi^*(x) - \phi_0(x) < 0$$

$\therefore$ $g(x) := (\phi^*(x) - \phi_0(x))(f_1(x) - K f_0(x)) > 0 \ \forall\, x \in B$

$= 0 \ \forall\, x \in B^c$

$= B_1 \cup B_2$

Suppose: $\mu(B) > 0$

$$\int_B g(x)\,dx > 0 \qquad \left[\text{if } f \geq 0, \int f\,d\mu = 0 \Rightarrow f = 0 \ \mu \text{ a.e.}\right]$$

$$\Rightarrow \int_B g(x)\,dx = \int_{\mathbb{R}^n} g(x)\,dx > 0$$

$$\Rightarrow \int (\phi^*(x) - \phi(x))(f_1(x) - K f_0(x))\,dx > 0$$

$$\Rightarrow \underbrace{\int (\phi^*(x) - \phi(x)) f_1(x)\,dx}_{\mathbb{E}_1 \phi^* - \mathbb{E}_1 \phi} > K \underbrace{\int (\phi^*(x) - \phi(x)) f_0(x)\,dx}_{\underset{=\alpha}{\mathbb{E}_0 \phi^*} - \underset{\leq \alpha}{\mathbb{E}_0 \phi}}$$

$\Rightarrow \mathbb{E}_1 \phi^* > \mathbb{E}_1 \phi$, contradiction!

$$\therefore \mu\{x : \phi^*(x) \neq \phi_0(x),\ f_1(x) \neq K f_0(x)\} = 0$$

$\Rightarrow$ either $\underbrace{\phi_0(x) = \phi^*(x)}_{\text{MP level } \alpha \text{ test } \phi_0 \text{ matches with } \phi^*}$ OR $\underbrace{f_1(x) = K f_0(x)}_{\phi_0 \text{ may not match with } \phi^* \text{ on the boundary}}$ μ a.e.

$\therefore$ If $f_1(x) > K f_0(x) \Rightarrow \phi_0(x) = \phi^*(x) = 1$

If $f_1(x) < K f_0(x) \Rightarrow \phi_0(x) = \phi^*(x) = 0$